

1 Research question and variables

The independent variable is the distance between the slit and the screen and will include values that range from the [nearby and further apart]. The [nearby] will include 20cm, 40cm, 60cm, 80cm, and 100cm while the further apart include 3 meters, 5 meters, 6 meters, and 8 meters.

Since the fringe spacing is within the millimeter range and it is beyond my human ability to hold the clipboard still within millimeter stillness while using the other hand to hold a pencil and mark accurately on the paper live. I decided to take a high resolution photo of the projection on a graphed piece of paper as a photo is most versatile and captures exactly as seen. The graphed piece of paper is printed with millimeter accuracy markings. as a result I will measure distance between each adjacent fringe (in millimeters) with software solutions and excuse my skill issue.

Research question. What is the relationship with the distance between screen and slit (from the range of 20cm to 8m) and the spacing between adjacent bright fringes (in mm) in a double slit system.

2 Background

Wave phenomena states that wave could interfere when they overlap. When two waves are in opposite phase, it is said to interfere destructively and cancel out. However, if they are the same phase and interfere, it is said to be constructive, reaching a greater amplitude.

The debate of whether light is a particle or a wave was challenged by Thomas Young's famous Double Slit Experiment in Young's Course of Lectures on Natural Philosophy and the Mechanical Arts of 1807, where a coherent light source shining through two tiny slits projected an interference pattern with alternation between light and dark fringes rather than two individual bright fringes in a dark background. The formula proposed by Thomas Young was

$$s = \frac{\lambda D}{d} \tag{1}$$

In which "S" is the distance between two adjacent bright fringes, λ is wavelength of the light, D is the distance between the screen and slits in meters and d is the distance between slits.

The mathematical undergirding is as thus justified[4]:

Engage thy eyes upon the following graph:

We observe that the path difference from the wave from the bottom slit and top slit is illustrated as Δx , this coincides with the equation

$$d \sin \theta = \Delta x \tag{2}$$

(pls Ms. Cupp make these equations smaller than x). We observe the fact that the path difference (Δx) for constructive waves is

$$n\lambda \tag{3}$$

where λ is the wavelength of the light and n is an integer representing the interested bright fringe.

3 Experimental design

Table 1: Variables, methods, and justifications

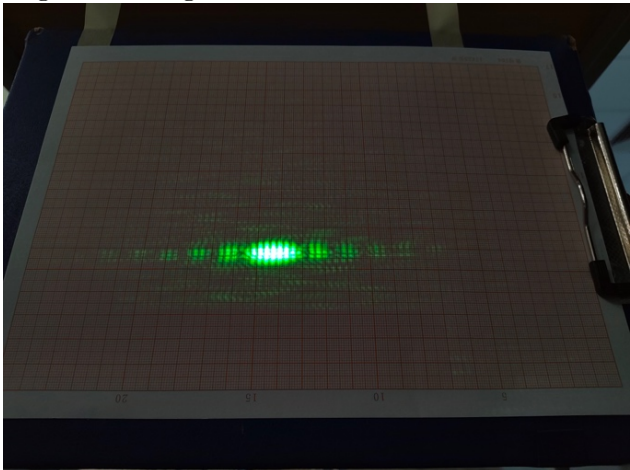
Independent variable	Method of how this variable will be changed	Justification: Why this range, interval or frequency was chosen?
----------------------	---	--

Distance between Slit and screen	For the long range, there will be three objects the pattern will be projected on, the object chosen is a graphed piece of paper on a clipboard, suspended hanging on a table. For each distance, the source will move between 3 distances, and once the data for each table is collected, it will be moved away to reveal the next table. For the short range data, the projected screen will move on a wheel on tracks.	This range of data will provide us justification for not only the short range, but also long range to show the validity of the double slit equation over a range of distances.
Dependent Variable	Method of how this variable will be measured	Justification: How many trials? How will precise and sufficient data be collected?
The distance between adjacent fringes on the projected screen	I will take a high resolution image with my phone and use perspective correction on the image and use software solutions such as ImageJ to find the average distance between each fringe	One picture will be taken for each distance. But multiple takes on fringe spacing will be recorded and the average will be found.
Control Variable	Potential impacts of this variable on the dependent variable/data	How it will be controlled and kept constant
The same laser and slit same method is used to measure the distances		

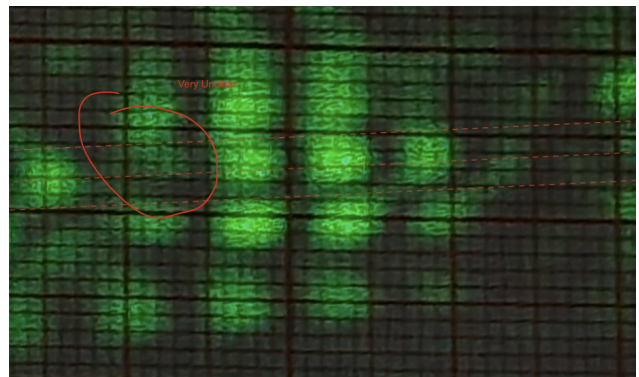
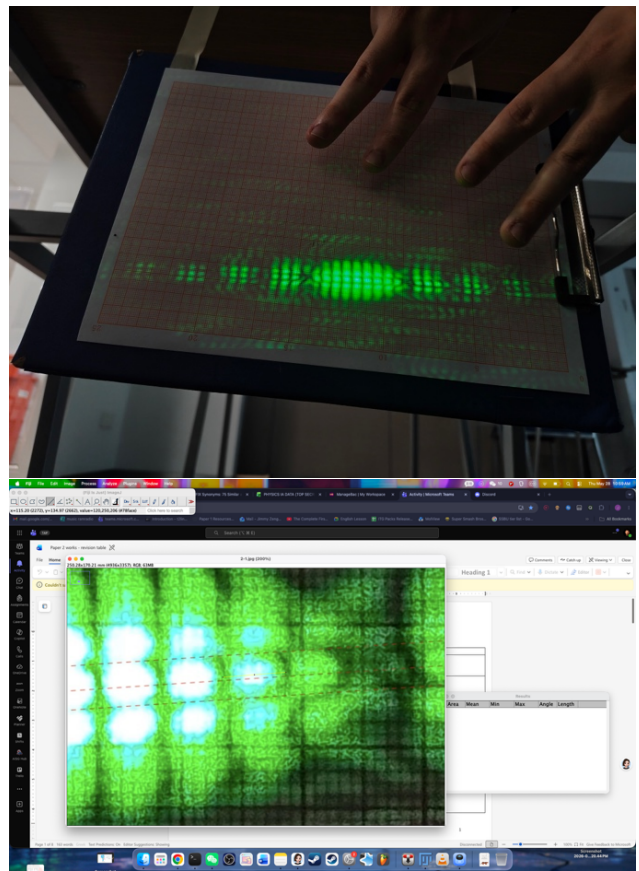
4 Data analysis

Qualitative: I don't recall having taped the paper onto the surface of the clipboard for maximum flatness of the surface

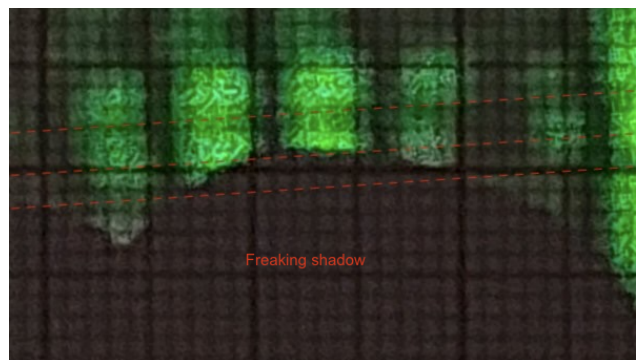
Table 2: Qualitative observations and interpretations

Qualitative observations	Interpretations/Photos
I don't recall having taped the paper onto the surface of the clipboard for maximum flatness of the surface. The paper is clipped on the right side but not the left	This means when we do the perspective shifts it might have slight distortions 

Some are really slanted for some reason. I ended up correcting it by drawing lines with respect to the direction of the slant and then measuring with that angle



Some of the fringes are too unclear to be seen I have noted them in my data when there were issues measuring



There are shadows sometimes in the data. This is also noted

5 Raw and processed measurements

Table 3: Measurements for data sets 1-1, 1-2, and 1-3

	1-1		1-2		1-3	
Pixels per cm square side length (pixels/cm). Used as a reference of size.	334.005	332.667	149.678	150.251	149.169	150.667
Pixels per cm square side length (pixels/cm). Used as a reference of size.	323.488	329.334	151.627	151.503	150.334	153.004
Pixels per cm square side length (pixels/cm). Used as a reference of size.	332.669	332.003	154.002	149.231	150.17	149.333
Mean	330.694		151.049		150.446	1667
Measurement of paper graph square side length after setting the mean (above) as reference	1.036	1.005	1.004	0.949	1.000	0.989
Measurement of paper graph square side length after setting the mean (above) as reference	1.013	0.953	1.081	1.01	0.975	0.972
Measurement of paper graph square side length after setting the mean (above) as reference	0.983		0.977		1.022	
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	Left edge	Right Edge	Left edge	Right Edge	Left edge	Right Edge
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	3.044	2.822	3.078	3.509	4.254	4.188
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	2.903	2.984	3.443	3.443	4.52	4.143
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	3.014	2.923	3.608	3.575	4.032	4.055
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	2.863	2.903	3.542	3.525	4.165	4.143

Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	2.963	2.903	3.542	3.426	4.099	4.099
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe	2.974	2.812	3.492	3.525	4.143	4.365
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe			3.343	3.525	4.055	4.032
Central Maxima (mm). Measured as the distance from the left/right edge of one bright fringe to the left/right edge of an adjacent bright fringe			3.542			
First Maxima (mm)	3.014	2.792	3.343	3.409	3.9	3.966
First Maxima (mm)	3.105	3.095	3.492	3.36	4.365	4.099
First Maxima (mm)	3.064	3.054	3.542	3.227	4.431	3.811
First Maxima (mm)	2.974	3.155	3.492	3.906	4.232	4.055
Second Maxima	2.933		3.426	3.062	4.032	3.877
Second Maxima	2.979		3.509	3.625	4.165	3.988
Second Maxima	2.903		3.294	3.707	4.032	4.032
(Data deemed difficult to measure consistently for the first one)	2.767		3.459	3.045	4.188	3.955
(Data deemed difficult to measure consistently for the first one)	3.22		3.261	3.327	4.276	4.154
(Data deemed difficult to measure consistently for the first one)	2.812		3.095	3.757	4.121	4.104
Average fringe displacement (mm)	2.96805		3.4416		4.119882353	
Expected/Calculated Value (mm)	2.9295		3.521		4.112	
Distance between laser and screen (cm)	247.8		297.8		347.8	
uncertainty of distance between adjacent bright fringes (mm)	± 0.2265		± 0.4305		± 0.3545	

Table 4: Measurements for data sets 2-1, 2-2, 2-3, and 3-1

2-1	2-2	2-3	3-1
198.668	247.335	236.513	162
197.333	249.668	237.334	162.507

196.667	198.669	248	247.337	235.675	237.51	162.501	161.757
196.667	195.336	248.002	247	225.281	229.847	157.009	161.174
197.223		247.89		233.693		161.158	
1.014	1.031	1.059	0.968	0.999	0.956	1.003	1.034
1.014	1.048	1.019	0.98	1.002	1.027	1.016	1.081
0.98		0.989		1.038		1.024	
Left edge	Right Edge	Left edge	Right Edge	Left edge	Right Edge	Left edge	Right Edge
4.91	4.322	6.102	5.866	6.994	6.262	6.144	6.408
5.542	5.137	5.922	5.883	6.39	6.342	6.227	6.557
5.306	5.316	6.099	5.74	6.524	6.393	6.455	6.186
5.129	5.219	5.839	5.901	6.436	6.601	6.599	6.517
5.287	5.215	6.042	6.042	6.479	6.61	6.683	6.417
5.479	5.413	5.987	5.9	6.302	6.308	6.372	6.31
5.186	5.262	5.984		6.479	6.732	6.537	6.682
5.108	5.371						
5.08	5.026	5.781	5.696	6.226	6.516	6.081	6.175
5.184	5.434	6.066	6.061	6.664	6.601	6.765	6.484
5.592	5.339	5.957	5.826	6.863	6.617	6.548	6.298
6.092	5.643	5.751	6.287	6.644	6.702	6.795	6.578
bad data		only clear data is extracted	only clear data is extracted			surprisingly consistent	surprisingly consistent
4.928		5.536	5.018	6.025	5.866	6.488	6.484
5.192		5.658	5.462	6.366	6.253	6.454	6.485
5.282		6.221	5.349	6.625	6.482		
4.834		5.723	5.933	6.177	6.869		
5.188		5.597	5.823	6.174			
5.386							
5.246733333		5.840387097		6.468451613		6.451115385	
5.294		5.885		6.476		6.476	
447.8		497.8		547.8		547.8	
± 0.885		± 0.6345		± 0.564		± 0.357	

Table 5: Measurements for data sets 3-2, 3-3, and 4-3

3-2		3-3		4-3	
155.766	156	189.667	191.167	162.675	163.672
154.253	154.001	192.667	191.707	162.003	160.502
154.002	157.851	195.668	185.783	162.5	163.773
155.312		191.11		162.521	
0.966	1.009	1.042	1.055	0.985	0.964
1.073	1.095	0.985	0.977	1.077	0.995
0.987		1.008		1.088	
Left edge	Right Edge	Left edge	Right Edge	Left edge	Right Edge
7.041	7.061	7.405	7.169	10.8	10.522
6.955	6.718	7.98	7.352	10.77	10.4
7.17	6.954	7.884	7.68	10.43	10.774
6.978	6.933	7.404	7.536	10.557	10.251
7.212	6.89	7.771	7.745	10.863	10.896
6.89	7.064	7.692	7.378	10.687	10.493

		7.536	7.956		10.585
			7.329		10.773
6.763	6.665	7.433	7.535	10.155	10.032
7.211	7.179	7.902	7.222	10.739	10.465
6.826	7.471	7.43	7.536	10.28	9.97
7.115	7.212	7.85	7.589	10.834	10.374
	data cropped out			Doesn't Exist	
7.052	6.632	7.589	7.614		
6.89	6.954	7.3	7.955		
6.993166667		7.584148148		10.52954545	
7.067		7.658		10.023	
597.8		647.8		847.8	
± 0.4195		± 0.4055		± 0.463	

Table 6: Short-range measurements at 100 cm, 80 cm, and 60 cm

100cm		80cm		60cm	
Chose to start with 100cm for mentality		too consistent and closeby additional measurement is deemed unnecessary			
146.503	145.255	199.167	199.379	226.002	226.003
145.333	145.668	199.506	199.878	225.938	226.256
146.51	146.168	199.357		226.168	226.295
145.906		199.4574		226.11	
Just as good as I needed it for this dis- tance :D					
1.017	1.005	1.019	1.036	0.984	1.021
1	1.057	1.019	1.028	0.999	1.058
1.028		1.103		1.028	

One error could be mid- dle ones reach max expo- sure on their edge and the fringes on the edge don't, but I measure the point of exposure change to max . This could also effect as we go out further	One error could be mid- dle ones reach max expo- sure on their edge and the fringes on the edge don't, but I measure the point of exposure change to max . This could also effect as we go out further		last one not measured	Very unclear	
Left Edge	Right edge	Left Edge	Right edge	Left Edge	Right edge
1.154	1.243	0.919	0.928	0.631	0.759
1.166	1.234	0.898	0.909	0.708	0.66
1.217	1.14	0.947	0.947	0.752	0.687
1.243	1.26	0.997	0.996	0.844	0.759
1.123	1.165	0.953	0.943	0.745	0.8
1.218	1.199	0.94	0.914	0.734	0.696
		0.915			
Very unclear					
1.062	1.115	0.953	0.821	0.756	0.747
1.303	1.15	1.028	0.953	0.708	0.763
1.097	1.226	1.009	1.028	0.807	0.636
1.08	1.131	0.984	0.936	0.829	0.73
1.1763		0.94847619		0.73755	
1.182222		0.945778		0.709	
100		80		60	
± 0.1205		± 0.1035		± 0.422	

Table 7: Short-range measurements at 40 cm and 20 cm

40cm		20cm	
208.417	208.5	232.417	231.75
207.667	208.375	232.855	232.629
208.625	209.188	231.5	231.878
208.462		232.172	
1.011	1.043	1.001	1.007
0.983	0.991	0.984	1.023
1.037		1.005	
Left edge	Right Edge	Left edge	Right Edge
0.496	0.424	0.319	0.287
0.468	0.512	0.287	0.266
0.552	0.552	0.276	0.291
0.488	0.504	0.259	0.226
0.512	0.5	0.255	0.291

0.476	0.552		
Able to see	Able to see	Able to see	Able to see
better with	better with	better with	better with
practice.	practice.	practice.	practice.
(Could've	(Could've	(Could've	(Could've
redone the	redone the	redone the	redone the
beginning	beginning	beginning	beginning
ones for	ones for	ones for	ones for
better ac-	better ac-	better ac-	better ac-
curacy, but	curacy, but	curacy, but	curacy, but
out of time	out of time	out of time	out of time
limitations)	limitations)	limitations)	limitations)
0.528	0.444	0.276	0.294
0.492	0.488	0.276	0.273
0.504	0.544	0.255	0.316
0.5	0.464	0.244	0.28
0.5		0.276166667	
0.473		0.236	
40		20	
± 0.064		± 0.0465	

6 Processing requirements and calculations

- Includes data tables with the proper units, number of sig figs (consistent decimal places) and uncertainties of measuring instruments.
- Uncertainties should be accurately recorded and processed.
- Includes related graph(s) with units, axis labels, error bars and any needed trend lines/slopes/R2 values.
- Includes a data table with slopes from the graphs and processed uncertainty
- All processing of data includes a short description along with example calculations of how data (including uncertainties) was propagated.

$$\text{Average fringe spacing} = \frac{\sum (\text{all fringe-spacing data points})}{\text{number of data points}} \quad (4)$$

$$\pm \text{Uncertainty} = \frac{\text{maximum value} - \text{minimum value}}{2} \quad (5)$$

7 Evaluation and conclusion

The goal of this research is to investigate in a double slit system, particularly how change in displacement between the slit and the screen corresponds to a change in [fringe spacing] and determine the curve relationship. In the data sheet I have included several notes as I was hand measuring the fringe by software tools, difficulties I encountered, anomalies from the norm or relevant observations.

In the beginning when I first measured the data, it was difficult to comprehend what should be the standard of what is the “edge” of each fringe, whether I should look at the image from the big picture and estimate where the image is implying it to be, or simply place it in the largest change in brightness. Because

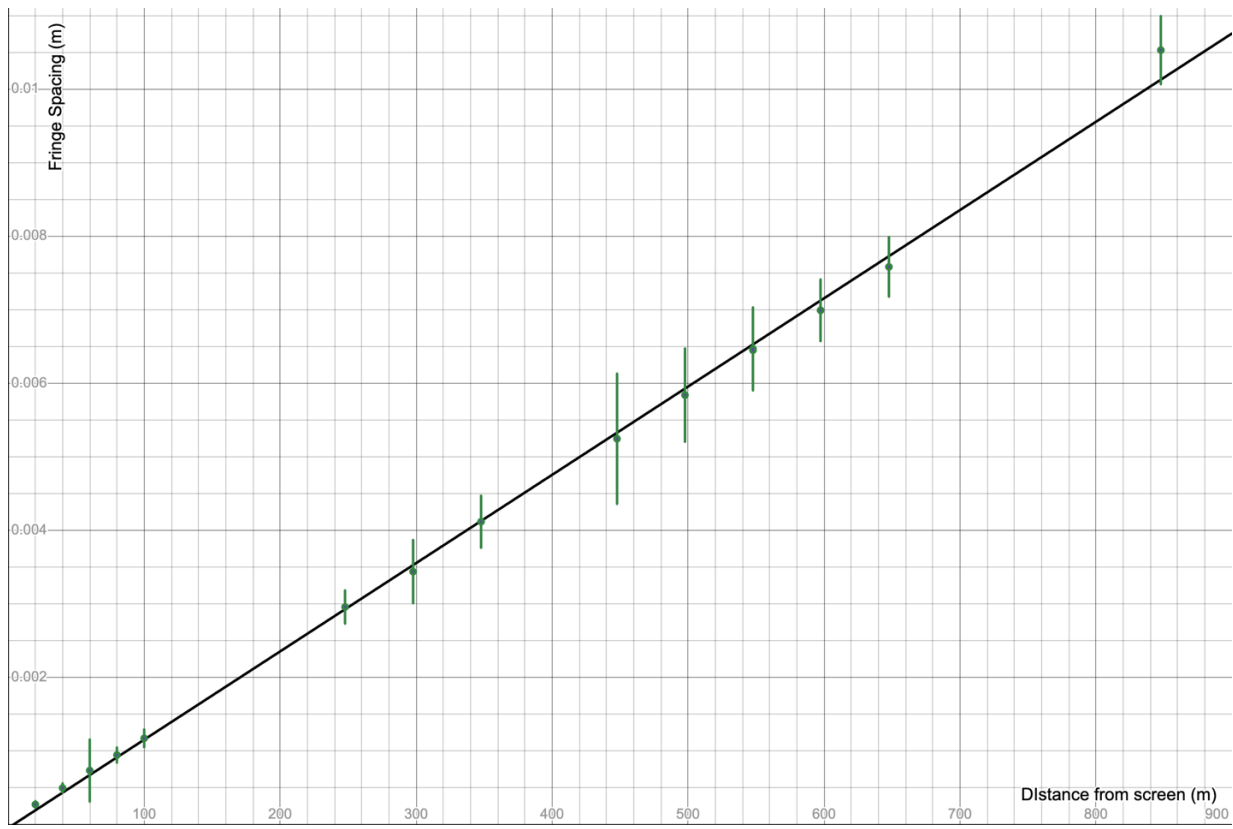


Figure 1: Processed-data graph with error bars and linear trend line.



Figure 2: Linear-regression output (equation and fit statistics shown in the source image).

if I copied the line to another edge-to-edge of the fringes, it looked as if that is where the image is implying the edge to be, though not explicitly described by the image. But for each image, the fringe spacing was rather consistent across all fringes, within the same mm digit, with slight variations in the tenths digit. When measuring from one image to the next, I can't help but notice a consistent increase of the fringe separation, indicating a linear relationship

We discovered that the relationship between these two variables is linear; an increase in the far/short-side displacement respects a [proportional increase] in fringe spacing, which validates the accuracy of Thomas Young's [formula] as the values measured are extremely similar to the calculated/expected values, up to micrometer accuracy, which I'm really proud of (please praise me).

A Limitations and proposed improvements

Table 8: Specific data issues, methodological weaknesses, and realistic improvements

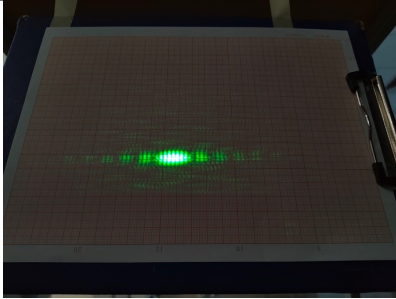
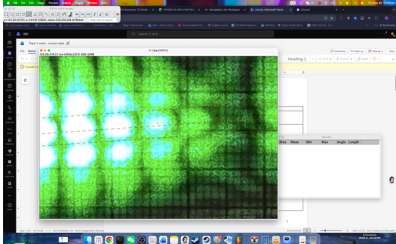
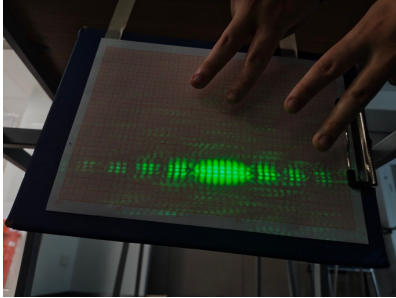
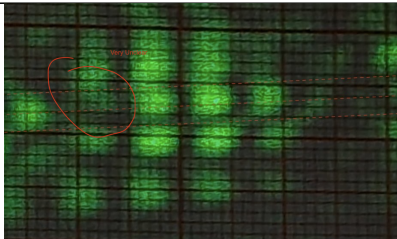
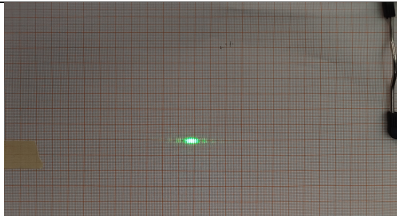
Specific Data (quoted from “Data Table 1” etc) that shows lack of precision or inaccuracy	Likely weakness in methodology or limitation that caused this lack of precision or inaccuracy in data	Realistic (could be completed in our high school lab) explanation of improvement
	For the far/short-sided approach, I didn't tape the paper onto the clipboard	This could lead to distortions in the perspective correction process as it is not inherently flat
	Some of the interference pattern isn't perfectly horizontal and are slanted	I had to draw lines that are parallel to the angle, and use that as a reference to accurately measure the distance in the software with greater validity
	Some of the lines I have drawn don't perfectly line up throughout the horizontal component of the interference pattern	I genuinely don't know why =

Table 9: Additional sources of uncertainty

	Some of the fringes are too unclear to be seen	I could spend more time adjusting camera properties and room lighting to ensure that the fringes are clear as well as the graphing paper's lines
	The fringes close to the middle are usually brighter, so it would more easily reach the maximum exposure of the camera	This can affect the perceived "edge" as that is my [baseline] for measuring fringe separation. I measure it by left/right edge to the edge of the adjacent fringe. This can compromise the data and introduce uncertainty. For example, when measuring distance between adjacent fringes of the leftmost edge, the left edge of that pair would be more close to the center than the right fringe of the pair, since this one is more brighter, the edge would be further from the center of the fringe.

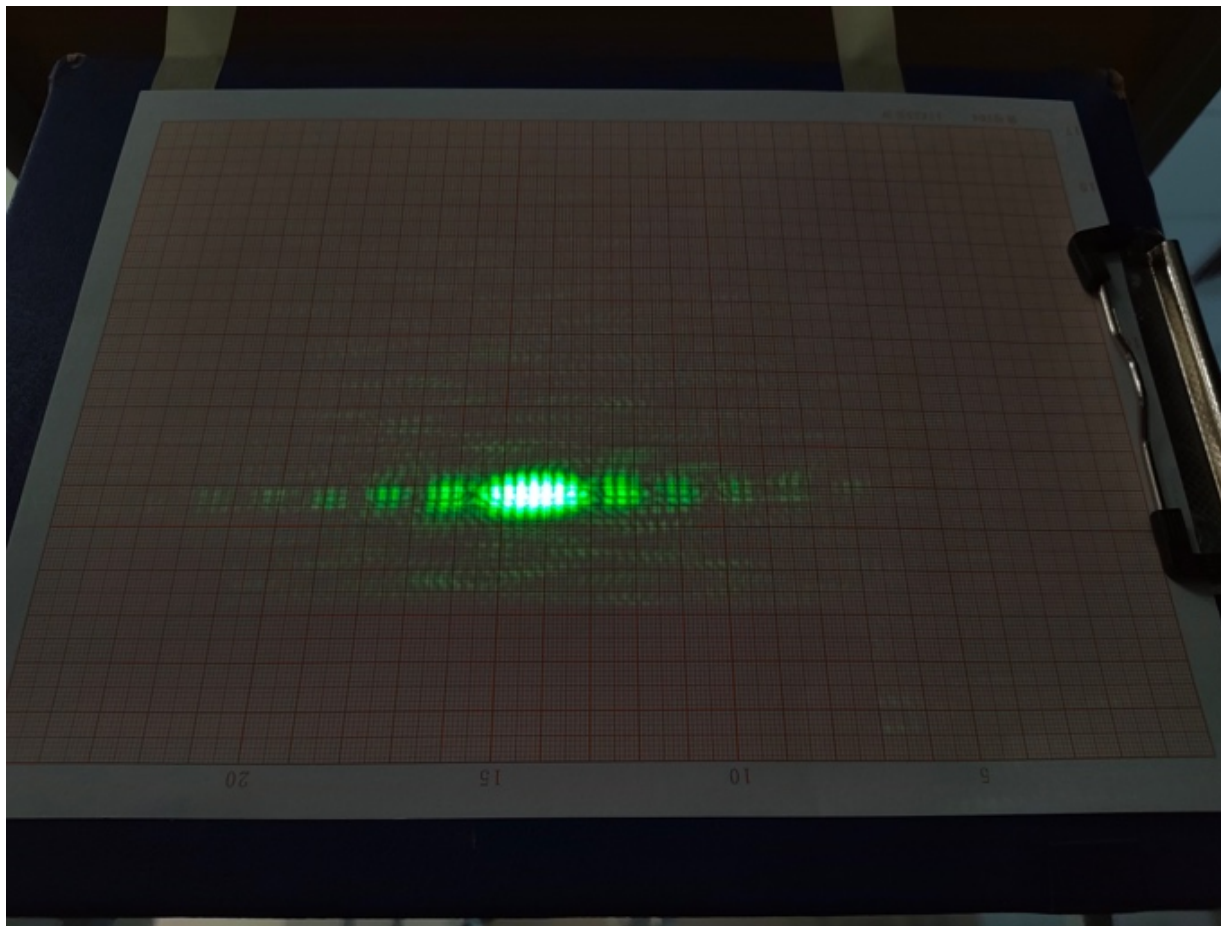


Figure 3: Original projection on graph paper (image 1).



Figure 4: Original projection on graph paper (image 2).

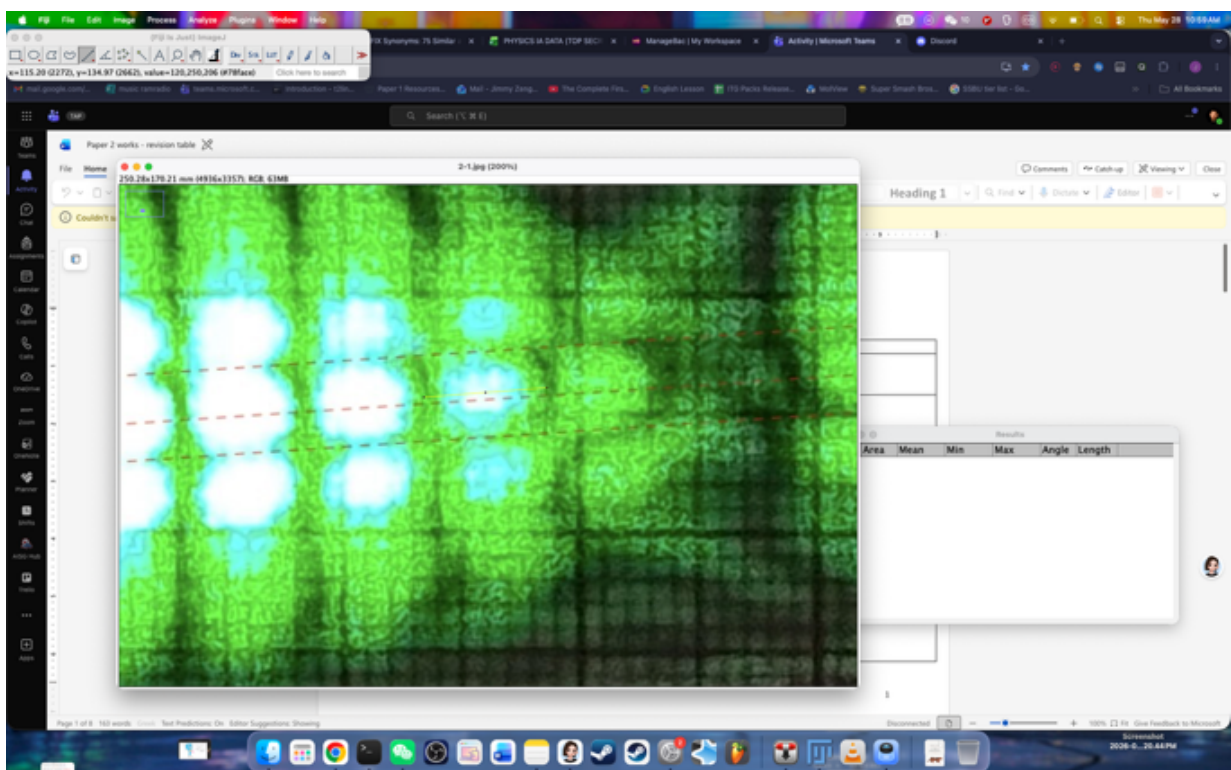


Figure 5: Software measurement view (image 3).

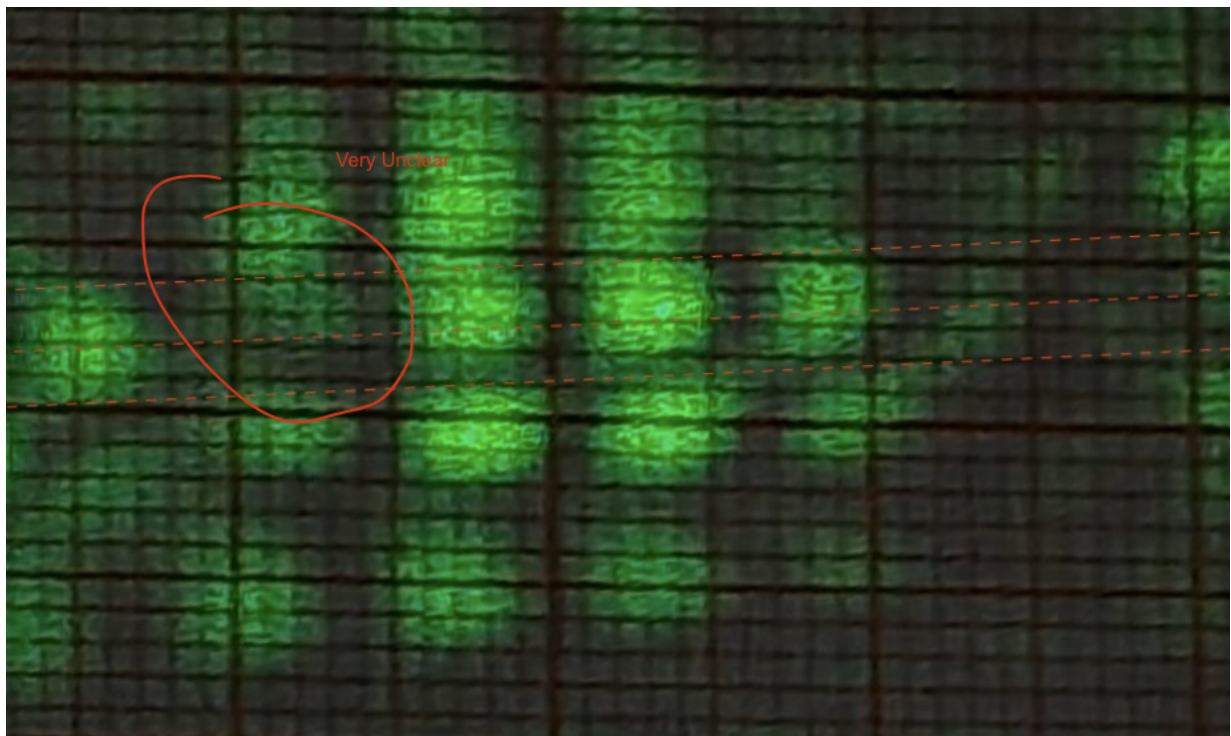


Figure 6: Interference pattern with marked region (image 4).

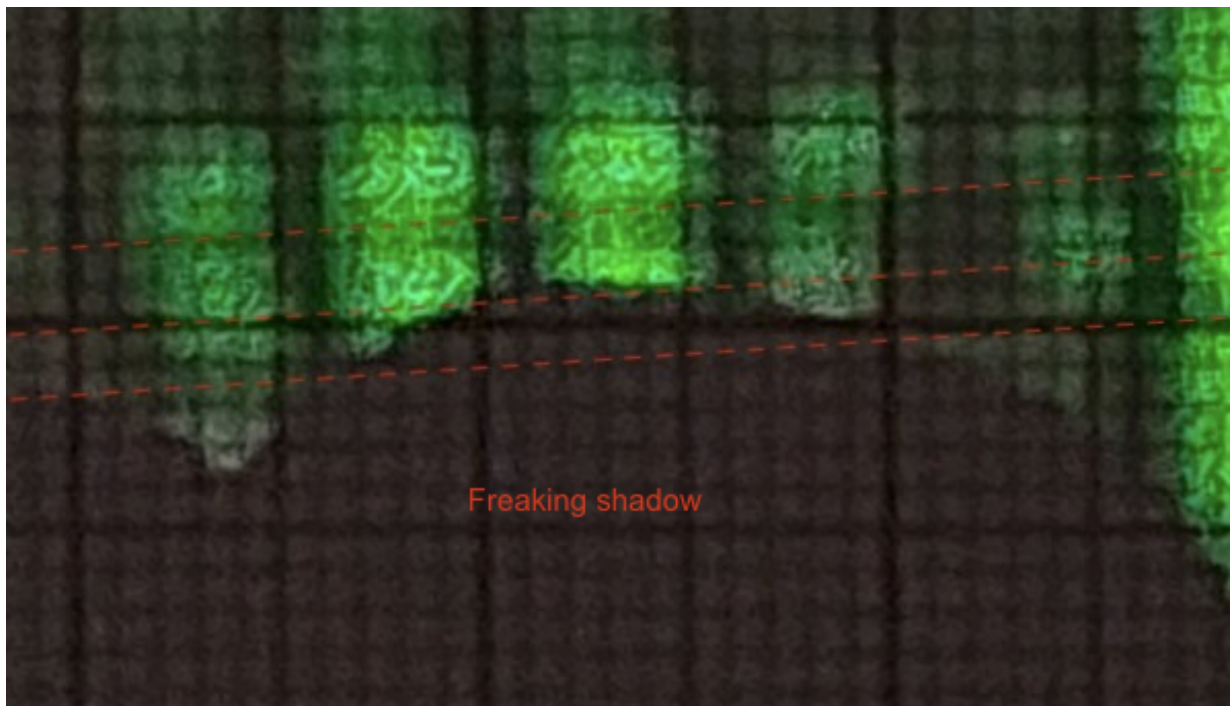


Figure 7: Interference pattern showing a shadow (image 5).

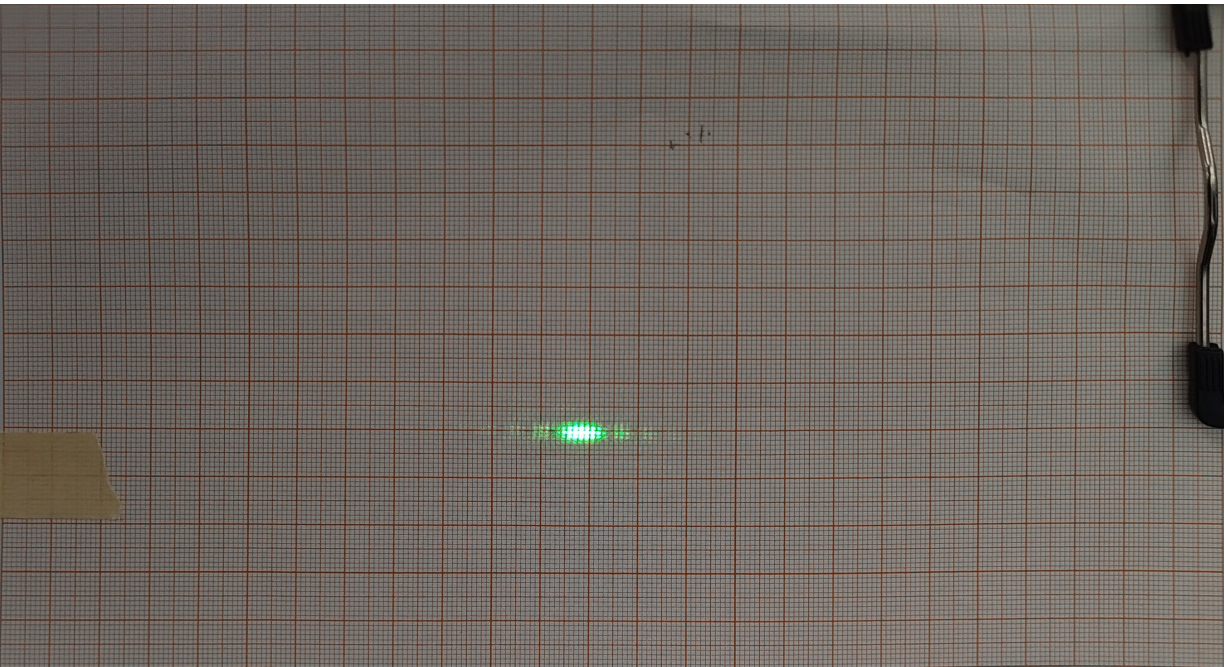


Figure 8: Long-range projection on graph paper (image 8).